Honors Calculus 1610

Lecture 15: Limits, Continuity, and the Differential

November 9, 2026

Today

- Limits and continuity when the input is a vector.
- A warning: approaching along every line is not enough.
- The differential $df|_{\vec{a}}$ of $f : \mathbb{R}^n \to \mathbb{R}$, carefully.
- The gradient $\nabla f(\vec{a})$, and how to compute it with partial derivatives.
- A sneak preview: differential forms (informal).

Limits with vector inputs

Let $D \subseteq \mathbb{R}^m$, let $\vec{f}\colon D \to \mathbb{R}^n$, and let $\vec{a} \in \mathbb{R}^m$. We want $\vec{f}$ to be defined at every $\vec{x}$ within some distance $r > 0$ of $\vec{a}$, though not necessarily at $\vec{x} = \vec{a}$:

$$\{\vec{x} : 0 < \|\vec{x} - \vec{a}\| < r\} \subseteq D.$$

Definition

We say $\lim_{\vec{x} \to \vec{a}} \vec{f}(\vec{x}) = \vec{L}$ to mean: for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$0 < \|\vec{x} - \vec{a}\| < \delta \quad \Longrightarrow \quad \|\vec{f}(\vec{x}) - \vec{L}\| < \epsilon.$$

Lecture 12 put lengths on the output. Now they are on the input as well. When $n = 1$, $\|f(\vec{x}) - L\| = \sqrt{(f(\vec{x}) - L)^2} = |f(\vec{x}) - L|$.

Example

Let $f : \mathbb{R}^n \to \mathbb{R}$ be $f(x_1, \ldots, x_n) = x_1$, and $\vec{a} = (1, 0, \ldots, 0)$. Claim: $\lim_{\vec{x} \to \vec{a}} f(\vec{x}) = 1$.

Given $\epsilon > 0$, let $\delta = \epsilon$. We want to check that $0 < \|\vec{x} - \vec{a}\| < \delta$ implies $|f(\vec{x}) - 1| < \epsilon$.

Since $\vec{x} - \vec{a} = (x_1 - 1, x_2, x_3, \ldots, x_n)$,

$$\delta > \|\vec{x} - \vec{a}\| = \sqrt{(x_1 - 1)^2 + x_2^2 + \cdots + x_n^2} \geq \sqrt{(x_1 - 1)^2} = |x_1 - 1| = |f(\vec{x}) - 1|.$$

So $|f(\vec{x}) - 1| < \delta = \epsilon$. □

Continuity

Definition

Let $\vec{f}\colon D \to \mathbb{R}^n$ be defined for all $\vec{x}$ close to $\vec{a} \in D$, that is, $\{\vec{x} : \|\vec{x} - \vec{a}\| < r\} \subseteq D$ for some $r > 0$. We say $\vec{f}$ is continuous at $\vec{x} = \vec{a}$ if $\lim_{\vec{x} \to \vec{a}} \vec{f}(\vec{x}) = \vec{f}(\vec{a})$.

On $\mathbb{R}$, x can approach a from only two sides.



In $\mathbb{R}^2$, $\vec{x}$ can approach $\vec{a}$ from every direction, along any path, inside a small ball.



Example / warning

$$f(x,y) = \begin{cases} \dfrac{x^2 y}{x^4 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Along the axes: $f(x,0) = 0/x^4 = 0$ and $f(0,y) = 0$. Along lines through the origin:

$$f(x,x) = \frac{x^3}{x^4 + x^2} = \frac{x}{x^2+1} \to 0, \qquad f(x,3x) = \frac{x^2(3x)}{x^4 + 9x^2} = \frac{3x}{x^2+9} \to 0,$$

and in the same way, for $m \neq 0$, $f(x,mx) = \dfrac{mx}{x^2+m^2} \to 0$ as $x \to 0$.

So along *every* line through the origin, f tends to $0 = f(0,0)$.

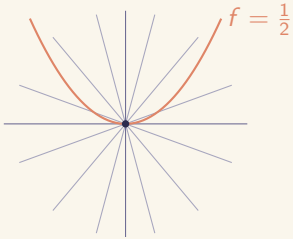
Along a parabola

Restrict to the parabola $y = x^2$: for $x \neq 0$,

$$f(x, x^2) = \frac{x^2 \, x^2}{x^4 + x^4} = \frac{x^4}{2x^4} = \frac{1}{2}.$$

So there are points arbitrarily close to the origin where $f = \frac{1}{2}$, and others (on the axes) where $f = 0$. The limit of f at the origin does not exist, and f is not continuous there.

(Along $y = x^3$, though, $f(x, x^3) = \frac{x^5}{x^4 + x^6} = \frac{x}{1 + x^2} \to 0$.)



Last time: covectors

Definition

A covector (a.k.a. linear function) is a function $m\colon \mathbb{R}^n \to \mathbb{R}$ such that

$$m(\vec{a}+\vec{b}) = m(\vec{a}) + m(\vec{b}) \qquad \text{and} \qquad m(\lambda\vec{a}) = \lambda\, m(\vec{a}).$$

Covectors $\longleftrightarrow$ vectors, via the dot product. Given a covector m, there is a unique vector $\vec{v}$ such that $m(\vec{a}) = \vec{v}\cdot\vec{a}$ for all $\vec{a}$.

Proof (existence). Given m, set $\vec{v} = (m(\vec{e}_1), m(\vec{e}_2), \ldots, m(\vec{e}_n))$. Writing $\vec{a} = a_1\vec{e}_1 + \cdots + a_n\vec{e}_n$,

$$m(\vec{a}) = m(a_1\vec{e}_1) + \cdots + m(a_n\vec{e}_n) = a_1 m(\vec{e}_1) + \cdots + a_n m(\vec{e}_n) = \vec{a}\cdot\vec{v}. \qquad \square$$

Three kinds of derivative

$f\colon \mathbb{R} \to \mathbb{R}$:

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}; \quad \text{i.e. } f'(a) = m \text{ means } \lim_{h\to 0} \frac{f(a+h) - f(a) - mh}{|h|} = 0.$$

$\vec{f}\colon \mathbb{R} \to \mathbb{R}^n$:

$$\vec{f}\,'(a) = \lim_{h\to 0} \frac{\vec{f}(a+h) - \vec{f}(a)}{h} = \lim_{h\to 0} \frac{1}{h}(\vec{f}(a+h) - \vec{f}(a)).$$

$f\colon \mathbb{R}^n \to \mathbb{R}$: $df|_{\vec{a}} = m$, a covector, with

$$\lim_{\vec{h}\to\vec{0}} \frac{f(\vec{a}+\vec{h}) - f(\vec{a}) - m(\vec{h})}{\|\vec{h}\|} = 0.$$

In one variable, $m(h) = mh$ is a very special linear function: multiplication by the number m.

The differential

Definition

The differential of $f : \mathbb{R}^n \to \mathbb{R}$ at $\vec{x} = \vec{a}$, denoted $df|_{\vec{a}}$, is the covector such that

$$\lim_{\vec{h} \to \vec{0}} \frac{f(\vec{a} + \vec{h}) - f(\vec{a}) - df|_{\vec{a}}(\vec{h})}{\|\vec{h}\|} = 0.$$

If $df|_{\vec{a}}$ exists with this property, we say that f is differentiable at $\vec{x} = \vec{a}$.

If f is differentiable at every point, then df assigns a covector to each point:

$$df : \ \vec{a} \longmapsto df|_{\vec{a}}, \qquad \mathbb{R}^n \longrightarrow \{\text{covectors}\},$$

a "covector-valued function".

The gradient: the vector version

Definition

The gradient of $f\colon \mathbb{R}^n \to \mathbb{R}$ at $\vec{x} = \vec{a}$ is the vector $\nabla f(\vec{a})$ such that

$$\lim_{\vec{h}\to\vec{0}} \frac{f(\vec{a}+\vec{h}) - f(\vec{a}) - \nabla f(\vec{a}) \cdot \vec{h}}{\|\vec{h}\|} = 0.$$

The covector $df|_{\vec{a}}$ and the vector $\nabla f(\vec{a})$ are partners, matched by the dot product:

$$df|_{\vec{a}}(\vec{h}) = \nabla f(\vec{a}) \cdot \vec{h}.$$

Remark. f has a differential at $\vec{x} = \vec{a}$ if and only if it has a gradient at $\vec{x} = \vec{a}$: every covector is $\vec{h} \mapsto \vec{v} \cdot \vec{h}$ for a unique $\vec{v}$, and $\vec{h} \mapsto \vec{v} \cdot \vec{h}$ is always a covector.

Computing the gradient

Suppose f has a gradient at $\vec{x} = \vec{a}$ (that is, f is differentiable there), and write $\vec{v} = \nabla f(\vec{a})$. Use only $\vec{h} = h\,\vec{e}_i$, with h a number: $\|h\vec{e}_i\| = |h|$ and $\vec{v} \cdot (h\vec{e}_i) = h\,v_i$, so

$$\lim_{h \to 0} \frac{f(\vec{a} + h\vec{e}_i) - f(\vec{a}) - h\,v_i}{|h|} = 0.$$

(The δ that works for $\vec{h} \to \vec{0}$ also works here: if $0 < |h| < \delta$ then $0 < \|h\vec{e}_i\| < \delta$.)

With $g(h) = f(\vec{a} + h\vec{e}_i)$, a function of one variable, this says $\lim_{h \to 0} \dfrac{g(h) - g(0) - h\,v_i}{|h|} = 0$: the second form of the derivative from Lecture 14. So $g'(0) = v_i$.

Partial derivatives

So the i-th component of the gradient is $(\nabla f(\vec{a}))_i = g'(0)$: the derivative of $f(\vec{x})$ at $\vec{x} = \vec{a}$, thinking of it only as a function of its i-th coordinate.

Definition

The *i-th partial derivative* of f at $\vec{a}$ is

$$\frac{\partial f}{\partial x_i}(\vec{a}) = \lim_{h \to 0} \frac{f(a_1, \ldots, a_{i-1}, a_i + h, a_{i+1}, \ldots, a_n) - f(a_1, \ldots, a_n)}{h}.$$

That is, if the gradient exists,

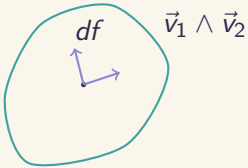
$$\nabla f(\vec{a}) = \left(\frac{\partial f}{\partial x_1}(\vec{a}), \frac{\partial f}{\partial x_2}(\vec{a}), \ldots, \frac{\partial f}{\partial x_n}(\vec{a}) \right), \qquad \nabla f \colon \mathbb{R}^n \to \mathbb{R}^n.$$

A sneak preview: differential forms (informal)

$C^\infty(\mathbb{R}^n)$: the "very" differentiable $f : \mathbb{R}^n \to \mathbb{R}$.

$$
\begin{array}{ll}
f \in C^\infty(\mathbb{R}^n) & \\
\downarrow d & (\sim \nabla) \\
df \in \Omega^1(\mathbb{R}^n) & \sim \text{vector-valued functions} \\
\\
\omega \in \Omega^1(\mathbb{R}^n) & \\
\downarrow d & (\sim \text{curl}) \\
d\omega \in \Omega^2(\mathbb{R}^n) & \sim \text{vector-valued, if } n = 3 \\
\downarrow d & (\sim \text{div}) \\
\Omega^3(\mathbb{R}^3) & \sim \text{real-valued functions}
\end{array}
$$

And $d(df) = 0$.



The general Stokes theorem:

$$\int_{\partial R} \omega = \int_R d\omega.$$